

HYDROTHERMAL CHEMISTRY AND GROWTH OF HEXAGONAL GERMANIUM DIOXIDE

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Hexagonal germanium dioxide GeO_2 is stable at $T > 1035^\circ\text{C}$. This modification of GeO_2 has theoretical and practical interest as a structural analog of α -quartz.

Three polymorphic modifications of crystalline GeO_2 are known: a low-temperature tetragonal modification with the rutile structure $\text{GeO}_{2(\text{tetr})}$, a high-temperature hexagonal form with the α -quartz structure $\text{GeO}_{2(\text{hex})}$, and a tetragonal modification with the low-temperature structure of cristobalite. The stability region of the last form has not been determined. The transition temperature $\text{GeO}_{2(\text{tetr})} \rightarrow \text{GeO}_{2(\text{hex})}$ is 1035°C at 0.1 MPa [1-3]. Two modifications exist under normal conditions: the tetragonal as a stable phase and the hexagonal as a metastable phase.

Two modifications can be prepared under hydrothermal conditions: GeO_2 with the rutile and α -quartz structures.

$\text{GeO}_{2(\text{hex})}$ has a 32 symmetry form and parameters which are similar to α -quartz:

α -quartz: $a = 4.903 \text{ \AA}$, $c = 5.393 \text{ \AA}$, space group $C3_12 (C3_22)$ [4];
 $\text{GeO}_{2(\text{hex})}$: $a = 4.987 \text{ \AA}$, $c = 5.652 \text{ \AA}$, space group $C3_12 (C3_22)$ [5].

However, α -quartz is a stable phase under normal conditions up to the α - β transition temperature of 573°C whereas the analogous GeO_2 modification is a metastable phase. The difference in chemical behavior of SiO_2 -quartz and of the GeO_2 quartz modification in water and aqueous solutions at room and elevated temperature has its origins here. This causes a significant difference for their growth conditions both with respect to temperature and solution compositions which can be used for single crystal growth.

1. BEHAVIOR OF GeO_2 IN AQUEOUS SOLUTIONS AT 25°C

1.1. Solubility of GeO_2 in Water at 25°C

A large number of publications are devoted to study of solubility and behavior of GeO_2 in water and aqueous solutions of acids and bases at room temperature. However, data on the state of GeO_2 in aqueous media are to a significant degree contradictory. Critical analysis of literature data and results of our investigations are examined in detail in [6, 7].

Results of experimental study of $\text{GeO}_{2(\text{hex})}$ solubility [4-20] in water and aqueous solutions at 25°C are converted by us to a single scale, $\log m_{\text{Ge}}$, and given in Fig. 1 (curve 1). It should be mentioned that the

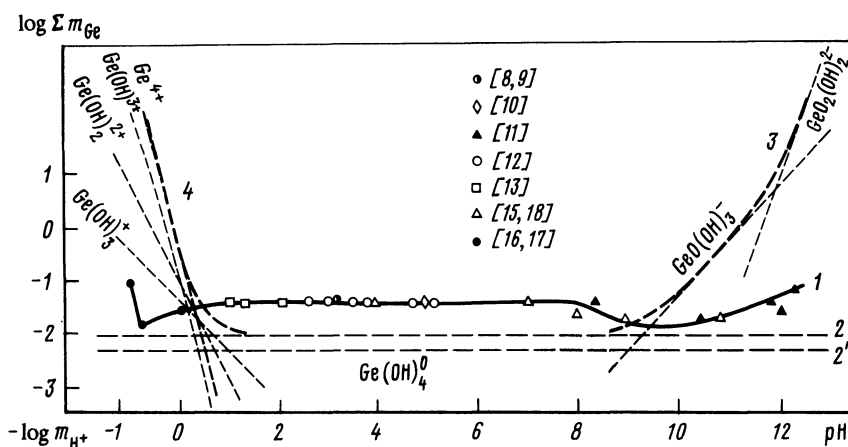
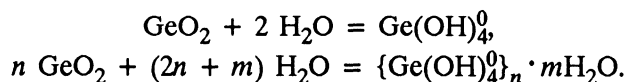


Fig. 1. Solubility of $\text{GeO}_{2(\text{hex})}$ at 25°C. 1) Experimental curve; 2) concentration of mononuclear Ge forms [24, 25]; 2') concentration of mononuclear Ge forms [14]; 3) theoretical curve of solubility under formation conditions of anionic forms of germanium in solution; 4) theoretical curve of solubility under formation conditions of cationic forms of germanium in solution.

equilibria attained during study of $\text{GeO}_{2(\text{hex})}$ solubility will be metastable since the stable solid phase is the rutile modification of GeO_2 . $\text{GeO}_{2(\text{tet})}$ besides $\text{GeO}_{2(\text{hex})}$ is observed as the solid phase in our experiments [7] even at 100°C with prolonged storage (20-30 days) in equilibrium with solution. The solubility of $\text{GeO}_{2(\text{hex})}$ in water at 25°C varies from $3.9 \cdot 10^{-2}$ to $4.8 \cdot 10^{-2} m$. We note that the solubility of $\text{GeO}_{2(\text{hex})}$ is two orders of magnitude higher than the solubility of α -quartz ($0.8 \cdot 10^{-4} m$) under analogous conditions and comparable to the solubility of amorphous silica [21].

The path of curve 1 in the pH 1-8 portion shows that the GeO_2 solubility does not depend on solution pH. This indicates that Ge transfers into solution as neutral particles* (monomeric and to a small degree polymeric):



Solubility determinations show that $\text{GeO}_{2(\text{hex})}$ solubility in both strongly acidic (pH < 1) and basic (pH 8-9) regions (Fig. 1, curve 1) is decreased. The GeO_2 solubility falls to a minimal value of 0.004 m (pH 9.1) with increased base concentration and then increases with further increase of pH.

A series of studies [9, 10, 15, 16] has shown that polygermanates corresponding to the cations $\text{M}_2\text{Ge}_5\text{O}_{11}$, $\text{M}_2\text{Ge}_5\text{O}_{11} \cdot n\text{H}_2\text{O}$, $\text{M}_2\text{Ge}_4\text{O}_9$, and others and not GeO_2 appear as the solid phase in equilibrium with the solution at pH above 8.5-9. The composition of these changes with increasing base concentration. Thus, curve 1 (Fig. 1) in the range pH ≥ 9 corresponds to the solubility of the corresponding polygermanates and not of GeO_2 .

An analogous change of $\text{GeO}_{2(\text{hex})}$ solubility is observed in the strongly acidic region (pH < 1) with increasing acid concentration. Experimental studies in solutions of hydrohalic acids [16-20, 22, 23] showed that

*The composition of these Ge-containing particles, depending on the coordination number of germanium in solution which by analogy with the solid phase can be equal to 4 or 6, can be described with a variable amount of water molecules, for example $\text{Ge(OH)}_4^0 \cdot 2\text{H}_2\text{O}$ or $\text{GeO(OH)}_3^- \cdot 2\text{H}_2\text{O}$. However, the formulas for the hydroxy-complexes given above will be used for further thermodynamic calculations since the free energy change of the hydration reaction in the standard state is equal to zero.

the $\text{GeO}_{2(\text{hex})}$ solubility curve is characterized by a minimum at an acid concentration of 4-6 *m*. The $\text{GeO}_{2(\text{hex})}$ solubility increases with further increase of acid concentration. This enhanced solubility is related in [18, 22] to formation of complexes such as $\text{Ge}(\text{OH})_x\text{Cl}_{6-x}^{2-}$ and $\text{Ge}(\text{OH})_x\text{Cl}_{5-x}^{-}$. A new solid phase, GeCl_4 [17], was discovered at $\text{HCl} > 8 \text{ m}$ in solubility experiments. Consequently, we do not have a "pure" $\text{GeO}_2\text{--H}_2\text{O}$ system even in the strongly acidic region.

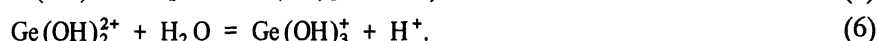
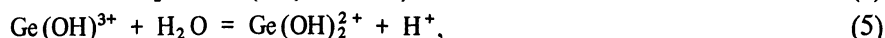
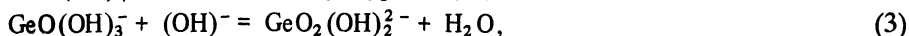
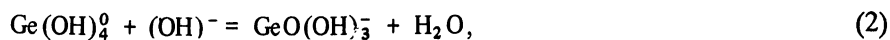
1.2. Forms of Germanium in Solution

Predominance of this or that form of germanium in solution in the absence of a complexing agent, as for a number of other cations such as Si, Al, and Sn, is determined by its total concentration Σm_{Ge} , the coordination number of germanium in solution, the concentration of H^+ (OH^-) ions, and the solution temperature.

A number of authors showed that monomeric germanium-containing particles dominate in solution at low total germanium concentration ($\Sigma m_{\text{Ge}} < 0.005 \text{ m}$ [24] or $\Sigma m_{\text{Ge}} < 0.01 \text{ m}$ [14, 24]) and in the absence of anion-complexing agents such as F^- , Cl^- , etc. These are monomers such as $\text{Ge}(\text{OH})_4^0$, $\text{GeO}(\text{OH})_3^-$, and $\text{GeO}_2(\text{OH})_2^{2-}$ [24, 25] in neutral and basic solutions (pH 7-13). Polynuclear particles, for example $\{[\text{Ge}(\text{OH})_4]_8(\text{OH})_3\}^{3-}$ [24], appear in solutions with germanium concentrations which exceed 0.005 *m* (0.01 *m*). The fraction of these increases with increased germanium concentration in solution. The logarithm of the formation reaction equilibrium constant of the $\{[\text{Ge}(\text{OH})_4]_8(\text{OH})_3\}^{3-}$ polyion is 29.14 ± 0.5 [24, 25]. Thus, it follows that the concentration of polyions with 3- charge at pH ≈ 4.3 and a total Ge concentration in solution above 0.005 *m* is 8 orders of magnitude lower than the monomer concentration, i.e., the concentration of charged polynuclear particles is negligible. The degree of polymerization of germanium particles depends on germanium concentration and solution pH.

The maximal Ge concentration in solution when all of the germanium is found as uncharged monomeric particles corresponds to curve 2 (or 2') (Fig. 1) according to [14, 24]. The experimentally determined curve lies above curve 2 (or 2'). This shows that uncharged polynuclear germanium particles besides the monomeric particles are also present in solution. The lowering of $\text{GeO}_{2(\text{hex})}$ solubility in the basic region up to the minimal value at pH 9.1 (NaOH concentration 0.004 *m*) attests to the disintegration of uncharged polynuclear particles to neutral $\text{Ge}(\text{OH})_4^0$ particles. Decreased $\text{GeO}_{2(\text{hex})}$ solubility in the acidic region up to the concentration of the $\text{Ge}(\text{OH})_4^0$ monomer (Fig. 1), as in the case of basic solutions, can be explained by depolymerization of Ge-containing particles.

Curves 3 and 4 (Fig. 1) correspond to the calculated change of GeO_2 solubility on pH if the GeO_2 dissolution occurs with formation of cationic and anionic forms according to the equations



The calculated curves 3 and 4 (Fig. 1) do not concur with the experimental GeO_2 solubility curve (curve 1). Since curve 1 in the pH > 9 region corresponds to polygermanate solubility and not to hexagonal GeO_2 , the right branch of the curve (Fig. 1, curve 3), which is constructed based on the formation reaction equilibrium constants of the $\text{GeO}(\text{OH})_3^-$ and $\text{GeO}_2(\text{OH})_2^{2-}$ complexes, pertains to metastable equilibrium in the $\text{GeO}_2\text{--H}_2\text{O}$ system. If it is accepted that germanium in the acidic region is present as the cations Ge^{4+} , $\text{Ge}(\text{OH})_2^{2+}$, $\text{Ge}(\text{OH})^{3+}$, and $\text{Ge}(\text{OH})_3^+$, in agreement with [26], then the solubility at pH < 1 should change according to

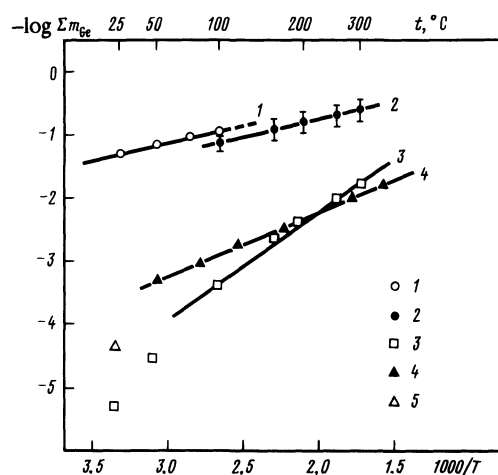


Fig. 2. Solubility of polymorphic modifications of GeO_2 in water as a function of temperature. 1) $\text{GeO}_{2(\text{hex})}$ (experimental solubility curve); 2) $\text{GeO}_{2(\text{hex})}$ (calculated curve); 3) $\text{GeO}_{2(\text{tetr})}$ (experimental solubility curve); 4) α -quartz from [21]; and 5) $\text{GeO}_{2(\text{tetr})}$ from [20].

curve 4 (Fig. 1). The path of the $\text{GeO}_{2(\text{hex})}$ solubility curve in this pH region shows that the data of [22] on the absence of cationic forms of germanium in solution should be considered most reliable.

Thus, cationic forms are absent both in the case of silicon (the $\text{SiO}_2\text{--H}_2\text{O}$ system) and for the $\text{GeO}_2\text{--H}_2\text{O}$ system. The difference between the germanium system and the $\text{SiO}_2\text{--H}_2\text{O}$ system, where silicon in the pH 1-9 range exists in the form of $\text{Si}(\text{OH})_4$ particles [21], is that neutral forms of germanium in solution can be both mononuclear and polynuclear. The increased solubility of $\text{GeO}_{2(\text{hex})}$ in comparison with quartz by two orders of magnitude (as a metastable phase) also explains the presence in solution of polynuclear forms of germanium.

2. BEHAVIOR OF GeO_2 AT ELEVATED TEMPERATURES

2.1. Solubility of GeO_2 in Water at Elevated Temperatures

The solubility of $\text{GeO}_{2(\text{hex})}$ in water (Fig. 2) between 25-100°C is described by the equation [7]

$$\log m_{\text{GeO}_2} = -\frac{767}{T} + 1.207. \quad (7)$$

Experiments on the determination of $\text{GeO}_{2(\text{hex})}$ solubility in H_2O at $T \gtrsim 120^\circ\text{C}$ showed that the obtained values of Ge concentration cannot be values of $\text{GeO}_{2(\text{hex})}$ solubility since the stable tetragonal GeO_2 phase was identified in experiments of more than 30 days duration. The quantity of $\text{GeO}_{2(\text{tetr})}$ increases with increased duration of the experiment.

We studied *in situ* the transition $\text{GeO}_{2(\text{hex})} \rightarrow \text{GeO}_{2(\text{tetr})}$ in water at 183°C using an energy-dispersive analysis which was described in [27]. The results are shown in Figs. 3 and 4. Lines of the $\text{GeO}_{2(\text{tetr})}$ spectrum are distinctly revealed at $T = 183^\circ\text{C}$ already 15 min after the initiation of heating. The starting $\text{GeO}_{2(\text{hex})}$ is completely converted into the tetragonal modification after 60 min. This fact suggests that the $185 \pm 5^\circ\text{C}$ temperature, proposed in the literature as the transition temperature for $\text{GeO}_{2(\text{hex})} \rightarrow \text{GeO}_{2(\text{tetr})}$ in the presence

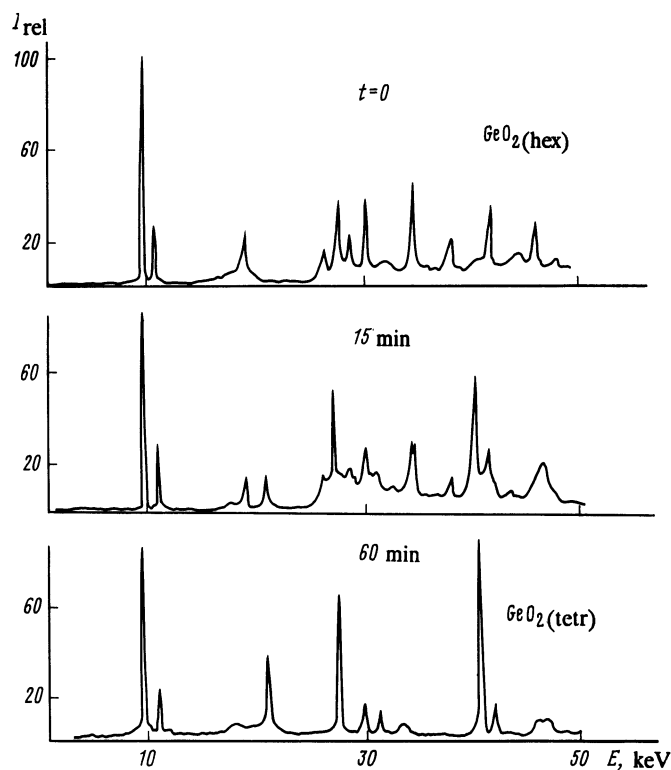


Fig. 3. Results of energy-dispersive analysis of the solid phase in the $\text{GeO}_2\text{-H}_2\text{O}$ system at 183°C .

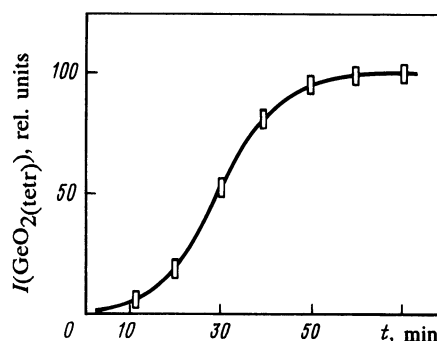


Fig. 4. Kinetics of the $\text{GeO}_{2(\text{hex})} \rightarrow \text{GeO}_{2(\text{tetr})}$ transition at 183°C from energy-dispersive analysis.

of water [28], is rather arbitrary. The transition of $\text{GeO}_{2(\text{hex})}$ into the stable $\text{GeO}_{2(\text{tetr})}$ modification is decidedly broadened and the reaction rate quickly increases with increasing temperature.

Thus, it seems impossible to determine experimentally the $\text{GeO}_{2(\text{hex})}$ solubility in water and aqueous solutions at $T > 120^\circ\text{C}$ (at lower temperatures the transition rate is sufficiently slow). It can be estimated from data on $\text{GeO}_{2(\text{tetr})}$ solubility and ΔG^0 as a function of temperature for the transition reaction $\text{GeO}_{2(\text{tetr})} \rightarrow \text{GeO}_{2(\text{hex})}$.

Solubility of the polymorphic GeO_2 modification with the rutile structure was studied by us [6, 7] between $25\text{-}300^\circ\text{C}$ at pressures from the saturated vapor pressure up to 140 MPa. The results are incorporated into Fig. 2.

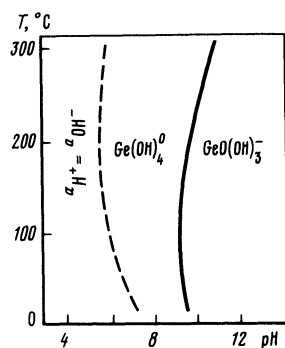


Fig. 5. Relation of hydroxo-complexes of germanium in aqueous solutions in neutral and basic regions as a function of temperature.

The experimentally determined value of $\text{GeO}_{2(\text{tetr})}$ solubility at 25°C is $4 \cdot 10^{-6} m$ [7]. This is approximately an order of magnitude lower than the value given in [19]. The $\text{GeO}_{2(\text{tetr})}$ solubility increases from $3.8 \cdot 10^{-4}$ to $1.7 \cdot 10^{-2} m$ with increasing temperature from 100 to 300°C. These values are substantially lower than the metastable solubility of $\text{GeO}_{2(\text{hex})}$.

Between 25-300°C, $\text{GeO}_{2(\text{tetr})}$ solubility is described by the equation

$$\log m_{\text{Ge}(\text{OH})_4^0} = \frac{-4132}{T} + 11.969 - 0.011559 T, \quad (8)$$

and between 100-300°C, by the linear equation

$$\log m_{\text{Ge}(\text{OH})_4^0} = -\frac{1774}{T} + 1.403. \quad (9)$$

Pressure in this temperature range practically does not affect $\text{GeO}_{2(\text{tetr})}$ solubility. Even at 300°C, when a change of the filling coefficient entails a sharp change of pressure (from the saturated vapor pressure up to 130-140 MPa), the obtained $\text{GeO}_{2(\text{tetr})}$ solubility values are identical within experimental error. The solubility does not depend on pH (as for $\text{GeO}_{2(\text{hex})}$ also) between pH 2-9 at constant temperature.

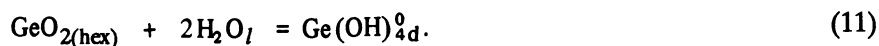
The solubility values of the stable α -quartz modification in water at these same temperatures [21] are shown in Fig. 2 for comparison. The order of magnitude of solubility of the stable α - SiO_2 modification is seen to be comparable to the solubility of the stable tetragonal modification of GeO_2 and not the isostructural but metastable $\text{GeO}_{2(\text{hex})}$.

The behavior of germanium hydroxo-complexes at various temperatures and solution pH values must be known in order to describe the equation of GeO_2 solubility under hydrothermal conditions. Estimation of reaction (3) equilibrium constants as functions of temperature allowed the behavior of germanium hydroxo-complexes at various temperatures and solution pH values to be determined [6]. Figure 5 shows that the field of dominance by germanium hydroxo-complexes depends weakly on temperature and that the mononuclear $\text{Ge}(\text{OH})_4^0$ particles dominate at elevated solution temperatures.

The $\text{GeO}_{2(\text{tetr})}$ solubility values are independent of pH, as shown by the fact that germanium transfers into solution under high-temperature conditions primarily as uncharged particles (Fig. 5). The low absolute value of solubility at 100-300°C ($< 0.01 m$) and the effect of temperature, which facilitates depolymerization processes, allow the $\text{GeO}_{2(\text{tetr})}$ dissolution reaction under the studied conditions to be described as



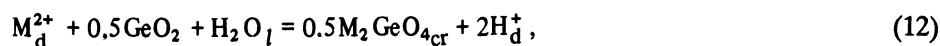
The $\text{GeO}_{2(\text{hex})}$ dissolution reaction, with a single mechanism for $\text{GeO}_{2(\text{tetr})}$ and $\text{GeO}_{2(\text{hex})}$ dissolution, can also be written as



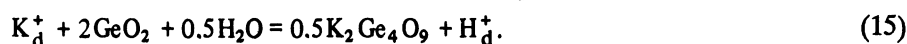
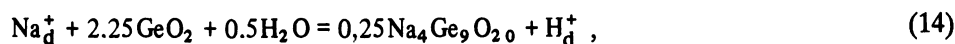
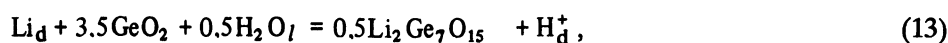
The solubility curve of $\text{GeO}_{2(\text{hex})}$ at $T > 100^\circ\text{C}$ (Fig. 2) is obtained from data on the $\text{GeO}_{2(\text{tetr})}$ solubility and the Gibbs free energy change for the transition $\text{GeO}_{2(\text{tetr})} \rightarrow \text{GeO}_{2(\text{hex})}$ as functions of temperature, calculated by us in [6]. The calculated curve, obtained accounting only for mononuclear forms of germanium in solution, lies lower than the experimental curve 1 (Fig. 2), i.e., part of the germanium in solution at $T > 25^\circ\text{C}$ is found as polynuclear particles.

2.2. Stability of GeO_2 in the Presence of Mono- and Divalent Cations

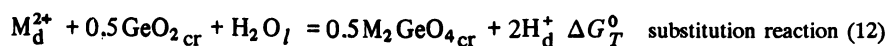
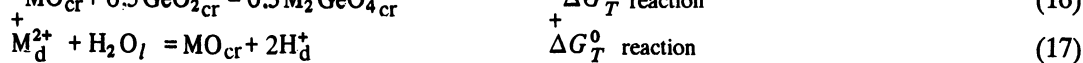
Growth of GeO_2 crystals will be limited by the kinetics of the $\text{GeO}_{2(\text{hex})} \rightarrow \text{GeO}_{2(\text{tetr})}$ transition, as follows from the arguments above. Besides this, the stability of GeO_2 depends substantially on the cationic composition and pH of the medium. Use of solvent mineralizers leads to replacement of GeO_2 by germanates. This was demonstrated above for $\text{pH} > 9$ and $T = 25^\circ\text{C}$. The stability of GeO_2 in the presence of mono- and divalent cations can be estimated using thermodynamic calculations. Equilibrium constants of the GeO_2 replacement reactions in aqueous solutions by the corresponding germanates must be considered for this. The germanate compositions were chosen in agreement with phase diagrams for $\text{GeO}_2\text{--MO}(\text{M}_2\text{O})$:



where $\text{M}^{2+} = \text{Ca}, \text{Zn}, \text{Fe}, \text{Co}, \text{Ni}$



Calculations were carried out for both $\text{GeO}_{2(\text{hex})}$ and $\text{GeO}_{2(\text{tetr})}$. The equilibrium constants for reactions (12)-(15) at various temperatures were calculated by us from the equations $\Delta G_T^0 = -RT \ln K$. The values of the Gibbs free energies ΔG_T^0 were found by summation of the free energies of the necessary reactions for which thermodynamic data are known



Equilibrium constants of reactions (13)-(15) for the corresponding alkaline germanates can be obtained analogously.

Standard formation enthalpies $\Delta H_{298.5}$, standard entropies $S_{298.5}^0$, and heat capacities as functions of temperature for the germanates examined must be known for calculating ΔG_T^0 of reaction (16).

Calorimetric data of [29, 30] were used for calculating the thermal change of reaction (16). These studies provide enthalpies of dissolution for the germanates Fe_2GeO_4 , Co_2GeO_4 , Ni_2GeO_4 , Zn_2GeO_4 , and Ca_2GeO_4 of the corresponding oxides of divalent metals and of $\text{GeO}_{2(\text{hex})}$ and $\text{GeO}_{2(\text{tetr})}$ in a melt of $2\text{PbO--B}_2\text{O}_3$ at 966 K . We calculated changes of enthalpy ΔH_{966} of reaction (16) from these measurements. The thermal effects of these reactions at 298.15 K were calculated with the assumption that the ΔC_p of reaction (16) as a function of temperature is the same as for the formation reaction from the oxides of the corresponding silicates. The silicate heat capacities as functions of temperature were taken from [31].

Calculation of the entropy change at 298.15 K in (16) necessitates availability of standard entropy values of the corresponding germanates. Values of the standard entropies of Li_2GeO_3 , Na_2GeO_3 , K_2GeO_3 , CaGeO_3 , and BaGeO_3 are given in [32]. Comparing these data with the standard entropies of silicates of analogous

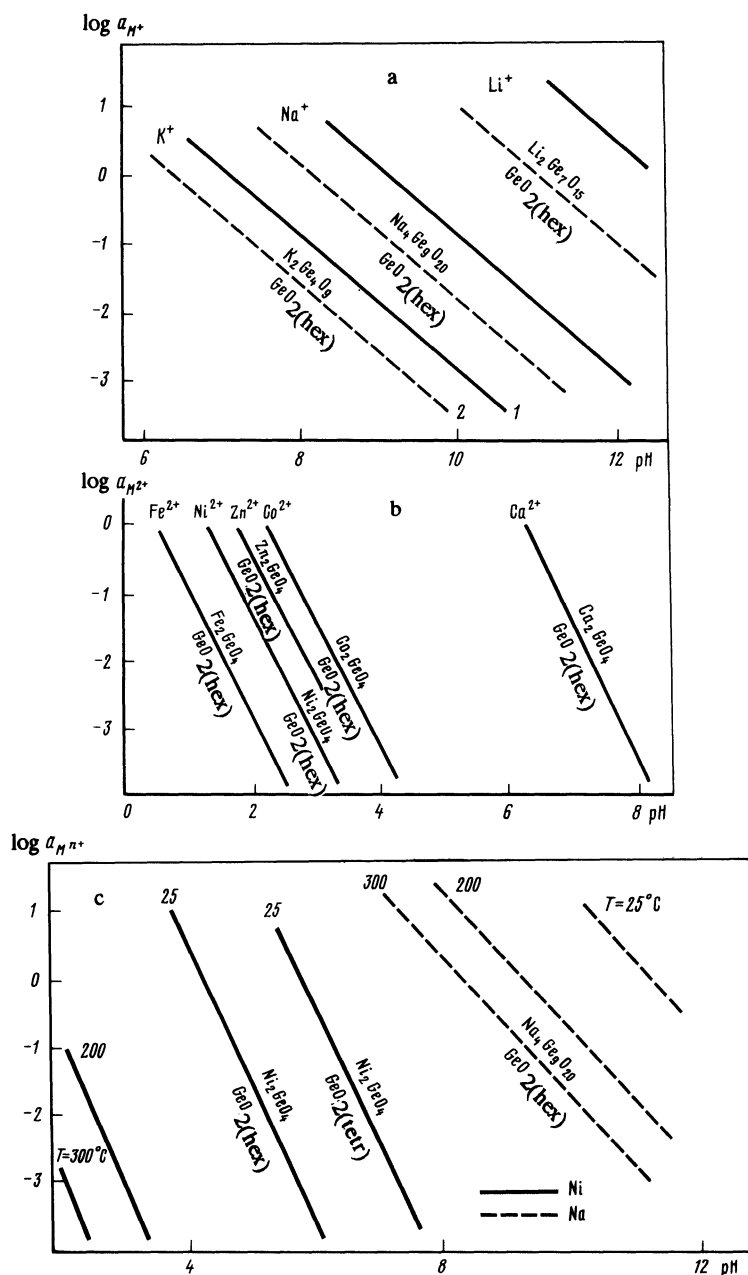


Fig. 6. Phase equilibria in systems based on GeO_2 . a) The $\text{Me}_2\text{O}-\text{GeO}_2-\text{H}_2\text{O}$ ($\text{Me} = \text{Li}, \text{Na}, \text{and K}$) systems at 200 (1) and 300°C (2); b) the $\text{MeO}-\text{GeO}_2-\text{H}_2\text{O}$ ($\text{Me} = \text{Ca}, \text{Fe}, \text{Ni}, \text{Zn}, \text{and Co}$) systems at 200°C; c) effect of temperature on phase equilibria in the $\text{Na}_2\text{O}-\text{GeO}_2-\text{H}_2\text{O}$, $\text{NiO}-\text{GeO}_2(\text{hex})-\text{H}_2\text{O}$, and $\text{NiO}-\text{GeO}_2(\text{tetr})-\text{H}_2\text{O}$ systems.

composition, we found that the relation between them can be expressed by the equation

$$S_{298,15}^0(\text{germanate}) = 5,071 + 0,915 S_{298,15}^0(\text{silicate}). \quad (18)$$

Coefficients of this equation were found by a least squares method for silicates and germanates of Li, Na, K, Ca, and Ba accounting for experimental error of the data. Equation (18) was used for estimation of the standard entropies of germanates $S_{298,15}^0$. The values obtained were 33.8 (Zn_2GeO_4), 36.8 (Fe_2GeO_4), 36.2 (Co_2GeO_4), and 29.4 (Ni_2GeO_4).

Free energy changes in the reactions (17) between 25-300°C were calculated according to [31].

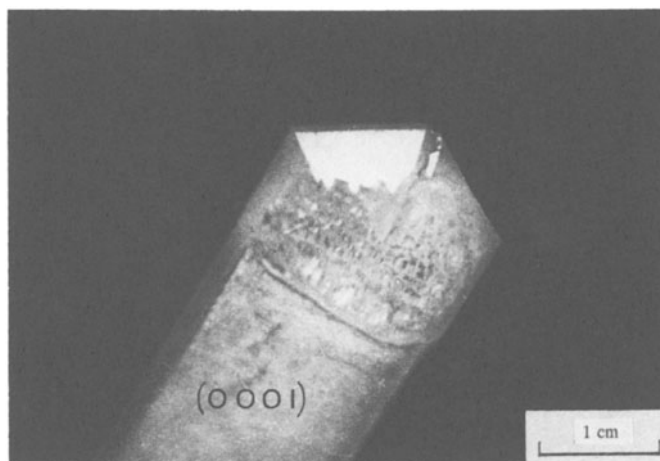


Fig. 7. Single crystal $\text{GeO}_{2(\text{hex})}$ grown on an α -quartz seed.

Results of calculations of phase equilibria in $\text{M}_2\text{O}(\text{MO})\text{--GeO}_2\text{--H}_2\text{O}$ systems between 25–300°C are given in Fig. 6a-c. The indicated tendencies hold for all the systems studied. $\text{GeO}_{2(\text{hex})}$ in the presence of alkali cations can be formed at temperatures below 200°C between pH 1–11 (at low cation activity). The stable field of GeO_2 is shortened with increasing ionic radius of the M^+ cation. The presence of K^+ ions leads to formation of germanates already in neutral and even weakly acidic solutions.

Introduction of divalent cations (Fig. 6b) leads to further shifting of the GeO_2 stability limits to more acidic solutions. The effect of hydrolytic processes on GeO_2 stability was evaluated for systems with divalent cations. Hydrolysis is important at low values of metal activities in solution ($\log a_{\text{M}^{2+}} \leq -2$ for Zn^{2+} and less than -4 and -5 for other divalent cations). The concentration of cations is usually substantially higher, $a_{\text{M}^{n+}} > 0$, than n moles in actual hydrothermal experiments on crystal growth. Therefore, hydrolytic processes can be neglected in this case. The $\text{M}_2\text{GeO}_4/\text{GeO}_2$ boundary accounting for hydrolysis is shown for Zn^{2+} in Fig. 6b.

The stability region of the stable GeO_2 tetragonal phase is wider than that of $\text{GeO}_{2(\text{hex})}$ in all cases in the presence of cations. This is demonstrated in Fig. 6c for the example of the system with Ni and Na. The increase of temperature also leads to a decrease of the stability field for GeO_2 .

The data obtained indicate that the region of possible $\text{GeO}_{2(\text{hex})}$ crystal growth is severely limited both in temperature and in solution pH value, as well as by the cationic composition of the medium. Growth of $\text{GeO}_{2(\text{hex})}$ is possible in weakly basic, neutral, or acidic solutions depending on the solvent. Stable germanates of basic divalent metals are formed in strongly basic solutions.

3. GROWTH OF $\text{GeO}_{2(\text{hex})}$ CRYSTALS UNDER HYDROTHERMAL CONDITIONS

We chose weakly acidic media — aqueous solutions of alkali metal fluorides and ammonium fluoride [33] — for growth of $\text{GeO}_{2(\text{hex})}$ crystals, in agreement with the examined features of $\text{GeO}_{2(\text{hex})}$ behavior in aqueous media at elevated temperatures. Preliminary experiments showed that $\text{GeO}_{2(\text{hex})}$ solubility in fluoride solutions practically does not differ from $\text{GeO}_{2(\text{hex})}$ solubility in pure water. An insignificant rise in solubility with increased concentration is observed in fluoride solutions, for example NH_4F from 0 to 1.5 mass %. With further increase of $C_{\text{NH}_4\text{F}}$, the solubility value is practically unchanged. The F^- ions which are added to the growth system should facilitate improved transport of germanium into the growth zone in comparison to pure water.

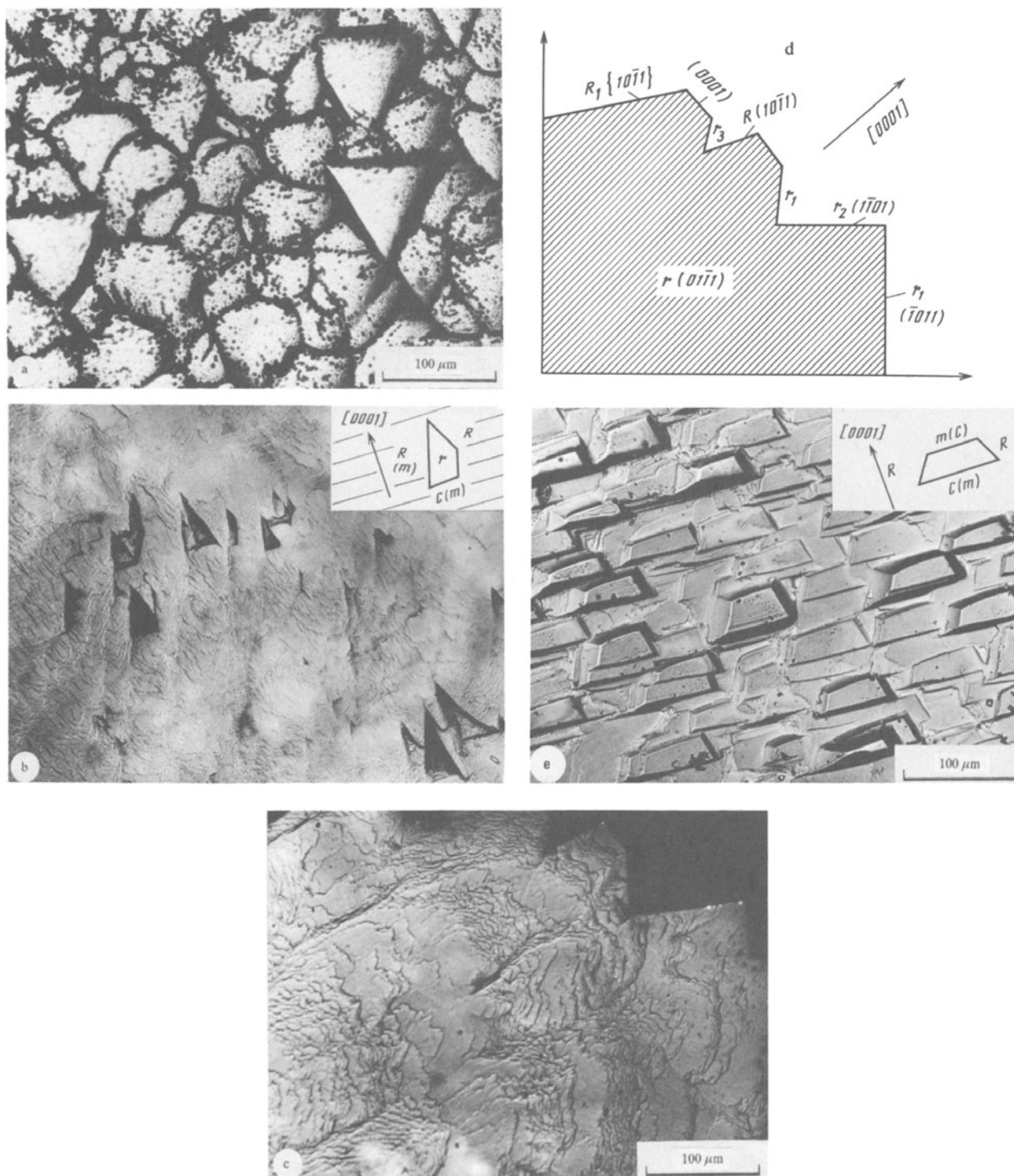


Fig. 8. Surface growth of GeO_2 crystal faces at small saturations. a) (0001) pinacoid face, V 0.04 mm/day; b) (0111) small rhombohedron r face, initial growth stage; c) (0111) small rhombohedron r face, cross section of growing pinacoid surface; d) scheme of limited mound growth on (0001) pinacoid face; e) (1011) large rhombohedron R face, V 0.01 mm/day.

The possible preparation of crystalline GeO_2 outgrowths on α -quartz in pure water was noted in [34]. The authors demonstrated a clearly elevated $\text{GeO}_{2(\text{hex})} \rightarrow \text{GeO}_{2(\text{tet})}$ transition temperature in the presence of water at 225°C. The layer that grew up to 2 mm in thickness was uneven and the quality was low.

The $\text{GeO}_{2(\text{hex})}$ was grown in 1-liter autoclaves fitted with floating fluoroplastic inserts. The temperature of the dissolution zone was 130-150°C. The temperature drop inside the autoclave was 1.5-7°C. The relatively

high temperatures were determined by the kinetics of the $\text{GeO}_{2(\text{hex})}$ crystallization process on a seed. They in turn determine the duration of the experiment, not more than 30 days, in order to avoid complete inversion of $\text{GeO}_{2(\text{hex})}$ into $\text{GeO}_{2(\text{tetr})}$.

Growth of $\text{GeO}_{2(\text{hex})}$ under metastable conditions was initiated by introduction of quartz substrate. Quartz plates which were cut parallel to the pinacoid faces (C -cut, $\{0001\}$), the $\{10\bar{1}1\}$ large R , and the $\{01\bar{1}1\}$ small r faces of the rhombohedra were used as seeds. Distilled water and solutions of ammonium, potassium, rubidium, and lithium fluorides were used as solvents.

Growth of GeO_2 was observed on quartz seeds of all these orientations. Significant difference in growth rates in various solvents was not observed. The crystallographic direction of growth and solvent type substantially affect adhesion of the grown layer. The maximal thickness of the grown layer was obtained on seeds perpendicular to the threefold axis (up to 5 mm, Fig. 7). The poorest growth was observed on seeds parallel to the $\{10\bar{1}1\}$ large rhombohedron. In this case, growth was practically not observed in NH_4F , H_2O , and KF solutions. A heteroepitaxial layer, which due to the heterometry and weak adhesion was easily separated mechanically from the substrate, was formed on the seed surface in LiF solutions. In contrast to quartz, as in [35], appreciable growth of the $\{10\bar{1}0\}$ hexagonal prism faces is observed. The growth rate of the prism faces exceeded the growth rate of the large rhombohedron faces. The relation of growth rates of $\text{GeO}_{2(\text{hex})}$ crystals in the solutions studied was: $V\{0001\} > V\{01\bar{1}1\} > V\{10\bar{1}0\}$, $V\{10\bar{1}1\}$.

The growth rate of $\{0001\}$ pinacoid faces reached 0.2 mm/day with ΔT inside the autoclave of about 7° . $V\{0001\}$ was ~ 0.04 and $V\{10\bar{1}1\}$ was ~ 0.01 mm/day for $\Delta T \approx 2^\circ$.

The morphology of the growing faces of GeO_2 crystals is different for crystals grown under low ($\sigma \approx 0.1$) and high ($\sigma \approx 0.5$) saturation conditions.

At low saturations ($\Delta T \approx 2^\circ$, $\sigma \approx 0.1$) and in the initial growth stages, the macrorelief of the growth layer surface is unique for each of the crystallographic substrate orientations studied. Figure 8a-e shows the surface growth for faces of GeO_2 crystals grown in lithium fluoride solutions on the C $\{0001\}$ pinacoid faces, the $\{01\bar{1}1\}$ small r , and the $\{10\bar{1}1\}$ large R faces of the rhombohedra. The orientational relations of the growth shapes on the various surfaces were determined by construction of the corresponding stereographic projections onto the faces.

Cellular growth, characterized by a specific "cobblestone" relief built primarily by rounded blocks, is observed on the pinacoid faces. Triangular blocks can be seen in a number of cases. The sides of the triangles correspond to cross sections of the planes by a particular type of rhombohedral face. The dark points in Fig. 8a are surface etching pits at the end of the growth experiment. The morphology of the pinacoid surface attests to the normal growth of the $\{0001\}$ faces.

Figure 8b shows the initial growth stage of the $\{01\bar{1}1\}$ small rhombohedron face. Growth occurs by layered growth. Traces of shifting of the pinacoid face (light hatching) are visible on the face. The growth front is perpendicular to $[0001]$. Stagnant portions which have a clear crystallographic orientation remain during shifting of layers. The sides of the negative growth shapes correspond to cross-sectional lines of the $\{01\bar{1}1\}$ R face and the faces of the pinacoid C (of the prism m)* and the $\{10\bar{1}1\}$ large rhombohedron R .

The portion of the face of the same crystal that is completely covered by the growing layer is shown in Fig. 8c. Expansion of the growth layers from the central portions of the face to the periphery is visible. The edges of the crystals are bounded by the planes of r - and R -rhomboheda in various combinations (Fig. 8c).

The surface of the large rhombohedron is covered by characteristic vicinal trapezoidal shapes that are regularly oriented on the face (Fig. 8d) and are formed by planes of the large and small rhombohedra and the pinacoid (of the prism m). The GeO_2 film peels in portions from the $\{10\bar{1}1\}$ R face.

*The face $C(m)$ belongs to the $\langle \bar{1}2\bar{1}0 \rangle$ zone, the face $R(m)$, to the $\langle \bar{1}\bar{1}23 \rangle$ zone.

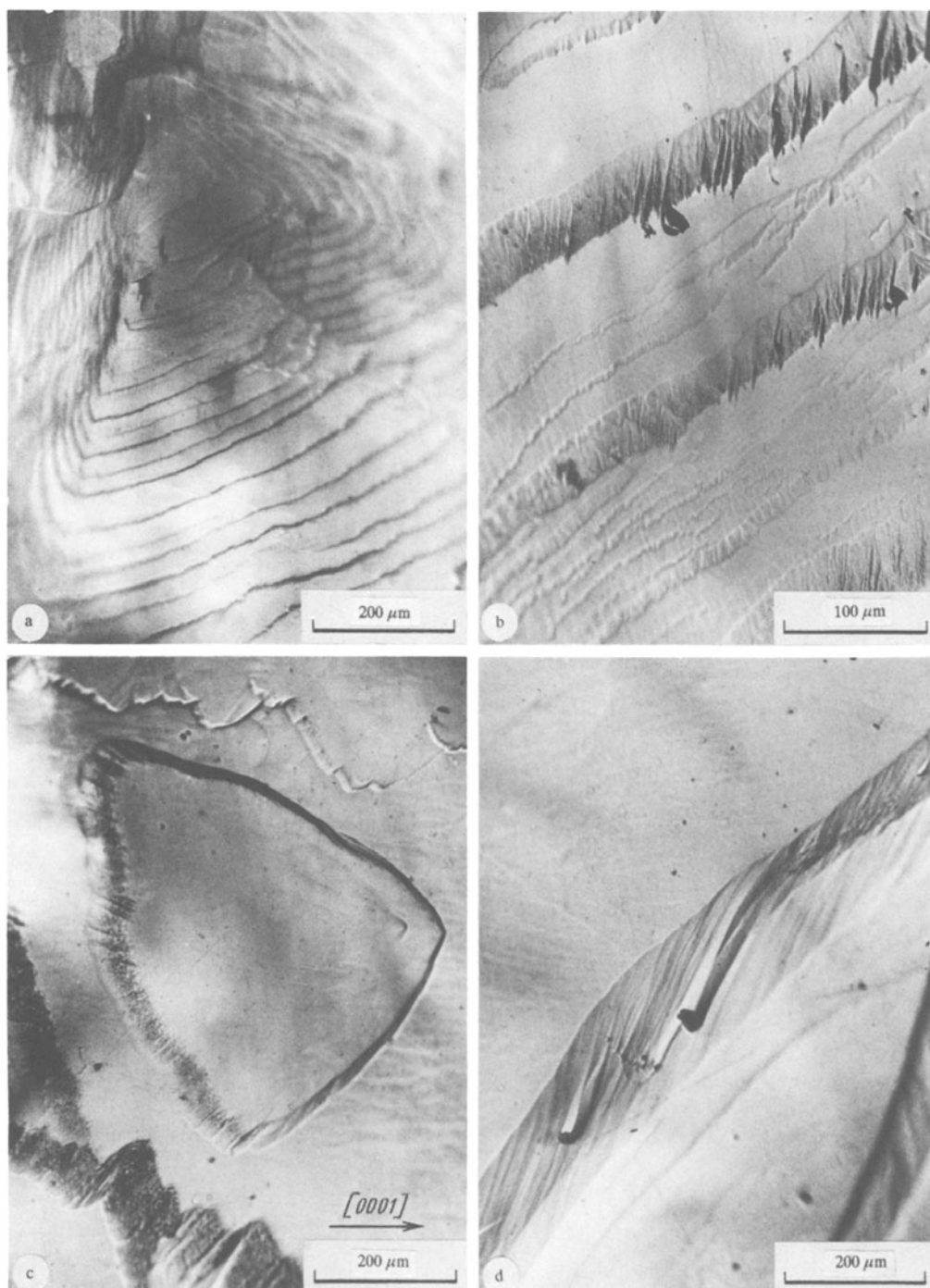


Fig. 9. Growth surface of GeO_2 crystal faces at large saturations. Spiral growth on the $(01\bar{1}1)$ rhombohedron face (a), characteristic view of growth steps on faces of rhombohedra and prisms (b), isolated step on the $(01\bar{1}1)$ rhombohedral face (c), and repulsion of mechanical impurities by the shifting face (d).

The morphology of the growth surfaces of GeO_2 crystals that are grown at large saturations often does not carry information on the crystallographic orientation of the grown surface. The morphology of the surfaces of the rhombohedra and prism is identical in general features. It is stepped and has rounded vicinal mounds (with macrospirals and without). The vicinal mounds with macrospirals (Fig. 9a) are usually located in the upper part of the crystal (the seed plates are positioned vertically). The density of steps is higher in the upper

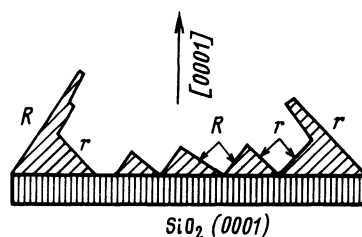


Fig. 10. Scheme of skeletal growth of GeO_2 on (0001) faces of SiO_2 .

part of the crystals, to which the current of most saturated solution flows, and gradually diminishes in the middle and lower portions of the face. The steps, which are curved in the middle portions of the face, bulging toward the lower side relative to the position in the autoclave, have assorted heights (Fig. 9b). The distance between high steps is on the order of $\sim 100 \mu\text{m}$. A series of steps of smaller width, $10\text{--}20 \mu\text{m}$, can be seen on the surface of these steps. Sometimes isolated islets are observed on the step surface. The surface of the step can be covered by characteristic rounded protruding mounds of growth (Fig. 9c). Repulsion of impurities during step advance can be seen in Fig. 9d. The GeO_2 growth mechanism on a quartz seed for the small rhombohedron and prism faces at low saturations can be considered layered. At large saturations, it is spirally layered.

Multinucleated growth is maintained on the pinacoid face. However, the relief of the face is not "cobblestone" but pointed triangular mounds that are bounded by rhombohedral r and R faces. Skeletal growth is often encountered at growth rates above 0.2 mm/day (Fig. 10).

4. CONCLUSIONS

1. The solubility of polymorphic modifications of GeO_2 in water at $25\text{--}300^\circ\text{C}$ is studied. The theoretical path of the solubility curve of the metastable modification $\text{GeO}_{2(\text{hex})}$ is calculated as a function of temperature. Dissolution of $\text{GeO}_{2(\text{hex})}$ in water and aqueous solutions at pH 1-8 occurs with formation of neutral, primarily mononuclear particles. Cationic forms of germanium are absent at pH < 1 . The stable solid phases at pH > 9 are germanates of alkali metals and ammonium.

2. The field of predominance of germanium hydroxo-complexes depends weakly on temperature. Uncharged germanium hydroxo-complexes predominate at elevated temperatures in neutral and weakly basic solutions.

3. Thermodynamic analysis of the stability of GeO_2 polymorphic modifications in ternary systems which contain alkali, alkaline earth, and transition elements showed that the stability of GeO_2 is also limited by the cationic composition of the medium. GeO_2 can be formed over a wide range of solution pH values (1-11). The field of its stability is shortened with increasing ionic radius of alkali metal (Li-Na-K) added to the system and with increasing temperature. Introduction of divalent cations leads to further shifting of the boundaries for GeO_2 formation to more acidic solutions. Stable alkali metal germanates crystallize from strongly basic solutions.

4. Growth of $\text{GeO}_{2(\text{hex})}$ under hydrothermal conditions in aqueous fluoride solutions on α -quartz seeds is studied. Rates of GeO_2 growth as a function of growth conditions are evaluated for various crystallographic directions. The morphology of the growing surfaces of GeO_2 crystals is studied.

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